University Library Use in Health Sciences and STEM Fields: An In-Depth Analysis of 2023-24 Survey Data from University of Michigan Faculty, Staff, and Students: Internship Final Report and Reflection

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Introduction

This project is an assessment of how faculty, staff, and students in the science, technology, engineering, and math (STEM) and health sciences fields use and perceive the services and resources offered by the University of Michigan (U-M) libraries. The findings were based on results obtained from participants from a survey conducted in the winter of 2023-24. The survey aimed to capture how these individuals experience the library to identify areas where the library can improve its services to better meet the needs of various academic communities. As the role of academic libraries in supporting research and education continues to evolve, understanding users' unique needs is critical for ensuring the library can effectively support the academic and professional pursuits of the U-M community.

Background

Due to various legal and statutory challenges, U-M has had to find creative ways to promote diversity, an essential component of academic excellence and institutional success. Affirmative action in admissions at U-M was first challenged in the Gratz v. Bollinger (2003) and Grutter v. Bollinger (2003) Supreme Court Cases. In the wake of these challenges, voters in Michigan banned affirmative action for public universities in 2006 (Blumberg & Weiland, 2022). As a result of these challenges, and the establishment of a new category for multiracial students, Black enrollment dropped from 7.1% in 2003 to

4.1% in 2015; in the same period, Native American enrollment dropped from 0.7% to 0.2% (U-M – Ann Arbor, 2003; U-M – Ann Arbor, 2015).

Since then, U-M has found ways to demonstrate its commitment to diversity. The university's Diversity, Equity, and Inclusion (DEI) strategic plan, initiated in 2016 and updated to version 2.0 in 2023, underscores a long-term commitment to creating a more equitable campus (Regents of the U-M, 2023).

Goals

When applying for the NCDS Data Internship, my primary goal was to gain hands-on experience in quantitative and qualitative data analysis methodologies and software utilization while collaborating closely with a team. I was also eager to learn about statistical analysis techniques beyond ANOVAs and crosstab reports. I was eager to work with data that required advanced techniques to account for real-world challenges like accounting for small sample sizes or comparing groups with vastly different numbers of respondents.

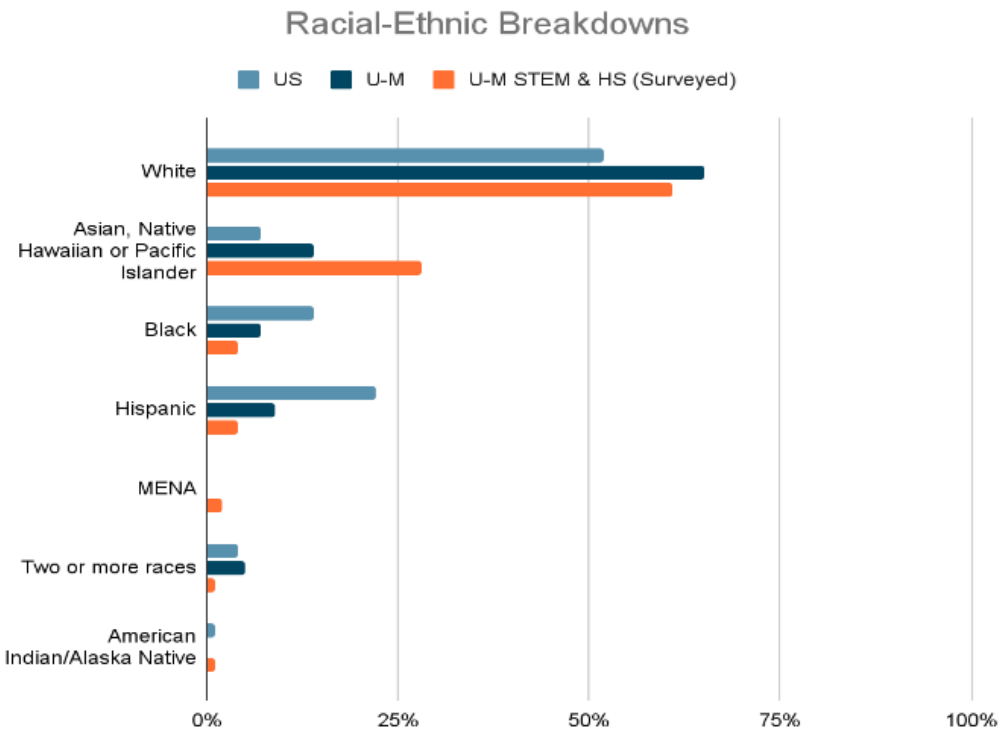
During the internship, each intern was able to choose a focus area. I decided that, in the wake of challenges to how U-M creates and promotes diversity on campus, I wanted to analyze the survey data to determine if and how diversity initiatives or the lack thereof have impacted how researchers, faculty, staff, and graduate students in the health sciences and STEM fields from different ethnic backgrounds use and perceive the library.

I hoped my contribution to the project could help U-M better understand factors influencing library use, identify barriers preventing some groups from utilizing these resources, and develop strategies to enhance communication and service offerings.

Additionally, I wanted to learn more about data cleaning, survey design, qualitative data coding, and various statistical techniques.

Outcomes

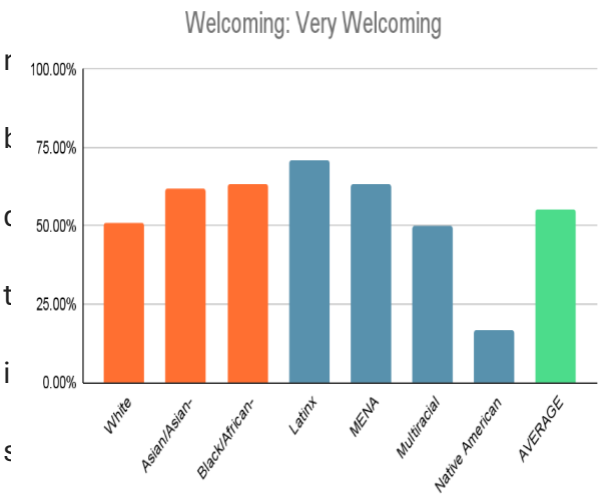
Focusing on students in the health sciences and STEM fields proved to be interesting but challenging. It was interesting because Black, Hispanic/Latine, and multiracial students were significantly underrepresented, while students from Asian backgrounds were overrepresented in the data; it was challenging because the low numbers of faculty, staff, or graduate students in health sciences or STEM fields who are Indigenous and Native American, Middle Eastern and North African (MENA), multiracial, Hispanic/Latine, and responded to the survey, made statistical analysis very difficult. I therefore concentrated my analysis on white, Asian, and Black respondents.

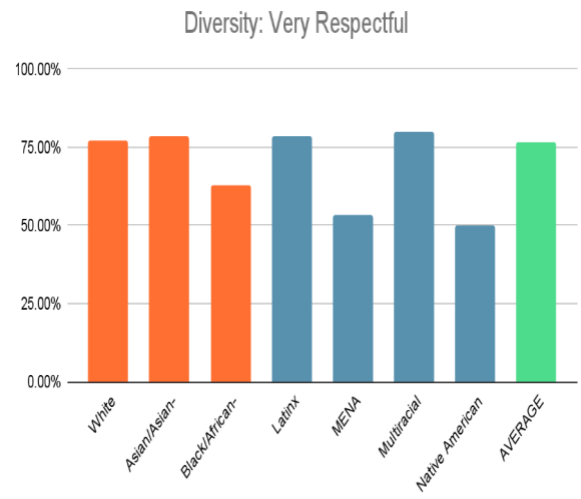


I analyzed responses for survey questions dealing with use and perception of the library, its collections, research support, and instruction support. My analysis found that there were no statistical differences in the way different groups used and perceived the library. The findings suggest that while the library is generally well-regarded, there are opportunities for improvement.

However, by digging deeper, I found some interesting differences. There were gender-based differences, with males generally feeling a higher sense of belonging and safety compared to females, particularly among Black/African-American respondents. I also found differences in the percentages of survey respondents from different ethnic backgrounds who gave the library the most positive ratings in certain categories. Two key findings were that: (1) a lower percentage of white respondents found the library very welcoming compared to Asian and Black students, and (2) a lower percentage of Black respondents felt that the library was very respectful toward diversity compared to white and Asian students.

These findings suggest some level of disen[illegible]ns. In[illegible]ackl[illegible]rs of[illegible]spect[illegible]aligr[illegible]





My suggestion for U-M is to follow the “Inclusion for All” framework developed by Brannon et al. (2018) to promote diversity and inclusion for the entire community *without* triggering reactance from dominant groups. Some members of majority groups may resist diversity initiatives due to a perception that these efforts limit their personal autonomy, sparking feelings of reactance. It's essential to incorporate opportunities for choice and autonomy within structured inclusion efforts to address this concern. Additionally, many individuals from dominant groups may prefer to maintain the status quo or adhere to a colorblind approach, avoiding discussions on race, which can result in resistance to diversity efforts. To mitigate this, it's crucial to create structured yet flexible environments that allow for intergroup interactions while ensuring that participants feel they have choices within the framework. Lastly, some may believe that racial and social equality has already been achieved, viewing diversity initiatives as unfair advantages for marginalized groups. Addressing this requires educating participants about historical injustices and their ongoing effects while emphasizing the fairness and legitimacy of diversity efforts, building a shared understanding of existing inequalities (Brannon et al., 2018).

Reflections and Lessons Learned

Managing my schedule this summer was more challenging than I anticipated, given my full-time job at the main location of my county’s public library system, unexpectedly needing to prepare and interview for a promotion, my part-time work as a Graduate Assistant, and my summer coursework. I had planned to front-load my internship work during the first four weeks, when my academic schedule was lighter and I was able to take some time off work. However, unanticipated delays and lack of preparation by the U-M

team hindered my ability to do this. Because the first team meeting was held two weeks into the internship and the software license for SPSS was unavailable until the fourth week, my ability to fully engage with the data analysis process was significantly impaired. Although eventually I reached out to NCDS and my teammate later reached out to our U-M team lead about the delay, I wish we had both reached out earlier. This experience taught me the importance of proactive communication and the need for better project management in collaborative settings.

I did very much appreciate the opportunity to practice interviewing for a data librarianship position. Although I wish I had received more feedback, the experience was incredibly valuable in helping me to prepare for the interview for the librarian position I currently hold. (My new position is not in data librarianship, so I will also be seeking a new role in data librarianship after receiving my MLIS.)

I am going to try to turn this experience into a positive one by presenting my findings at a conference with my fellow interns, and by writing up my findings for the South Carolina Libraries Journal with the support of my mentor, which will allow me to share the insights I gained and contribute to the broader academic community. Now that I have a basic familiarity with the data and some of the software, I want to continue to see what I can learn from working on this dataset, but I am not sure whether I want to continue working with the U-M team.

Areas of Improvement

In reflecting on areas of improvement, I found several issues that hindered the effectiveness of this internship experience. First, the delayed start of the project, due to

the late scheduling of the initial team meeting, indicates the need to better plan a schedule and coordinate long before the interns' official start date. Ensuring that all of necessary tools, such as software licenses, are ready before the project begins would also prevent delays. Additionally, having clearly defined learning objectives and a structured timeline from the start would help keep the team focused and ensure that all participants gain valuable skills and experience. "Create a schedule" was an agenda item four weeks in, which is unacceptable. Because of the condensed timeline of the internship, not enough time or attention was devoted to reviewing and analyzing existing literature. Finally, work must be done to rectify inconsistencies in communication and lack of follow-through on promised resources (such as access to Dedoose) and support (such as 1:1 meeting scheduling).

I was also disappointed that the bulk of the survey analysis work had been completed before the NCDS interns were added to the team. NCDS interns on other teams learned important data management skills that included cleaning and preparing data for analysis, coding data, and creating project-related documentation, but all of this work, plus a preliminary data analysis, had been completed by U-M before the NCDS internship began. I would suggest that, as part of their application materials, sites should submit a detailed timeline including tasks and learning goals, to NCDS. Further, NCDS should ensure that each learning goal contributes to the development of hands-on skills.

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